

## 7. Orwell-Vernon Estuary

Shoreline length: 71 km



### Shore Types

Orwell-Vernon Estuary is an estuary located on the southeast end of PEI. Shore type characterized by a majority of wetlands and low plains.

% Shore types within SMU

| low plain | bluff | sloped armour | cliff | wetland | vertical struct. | dunes | sandspit |
|-----------|-------|---------------|-------|---------|------------------|-------|----------|
| 21        | 4     | 2             | 3     | 70      | 0                | 0     | 0        |



Cliffs at McInnis Point



TCH at Vernon Bridge



Brush wharf



Low plain at the mouth of Vernon River estuary

## Governance, Infrastructure, and Economy

This area is unincorporated. Key infrastructure includes Orwell Corner Historic Village, Earnscliffe Road (Route 267, at the beach); Route 270 (McInnis Point Road, at a sharp turn, with potential to isolate portions of McInnis Point); Route 268 (China Point Road, at Watts and China Creeks); the TCH at Vernon Bridge; Murray Harbour Road at Vernon River; the TCH at Orwell (Uigg Road); Brush Wharf Road (Orwell Cove); and Frand Road (Orwell). Coastal economic drivers include Bill & Stanley's Oyster Company at Brush Wharf Road. Local and legacy land use constraints include several small lots, accessed by unpaved coastal roads.

## Natural Assets and Heritage

The Orwell Corner Historic Village is found at the northernmost point of this SMU. Additionally, the Sir Andrew Macphail Homestead is located in this SMU.

## Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

## Sediment transport

The estimated net sediment transport rate averaged across the SMU is low, except for the exposed outer cliffs near McInnis Point.

## Coastal Flood and Erosion Risk

Infrastructure along low-lying shores are vulnerable to increasing flood hazards. Erosion hazard is limited due to the sheltered orientation of the area.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

|           | FLOOD RISK     |                                  |                                   |                                 | EROSION RISK          |                        |                      |
|-----------|----------------|----------------------------------|-----------------------------------|---------------------------------|-----------------------|------------------------|----------------------|
|           | Present        | Short-term<br>0.3m SLR<br>(2050) | Medium-term<br>0.6m SLR<br>(2080) | Long-term<br>1.2m SLR<br>(2130) | Short-term<br>(~2050) | Medium-term<br>(~2080) | Long-term<br>(~2130) |
| Buildings | 2 [<1/km]      | 2 [<1/km]                        | 6 [<1/km]                         | 7 [<1/km]                       | 2 [<1/km]             | 4 [<1/km]              | 11 [<1/km]           |
| Roads     | 154 m [2 m/km] | 227 m [3 m/km]                   | 366 m [5 m/km]                    | 515 m [7 m/km]                  | 0                     | 0                      | 220 m [3 m/km]       |

## Proposed SMP Strategies

|                                   | Short-term                                                                                                                                                                                                                                                                                                                             | Medium-term | Long-term   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| Sea level rise range              | 0 – 0.3 m                                                                                                                                                                                                                                                                                                                              | 0.3 – 0.6 m | 0.6 – 1.2 m |
| Potential timeframe               | 2030 - 2050                                                                                                                                                                                                                                                                                                                            | 2050 - 2080 | 2080 - 2130 |
| <b>7.00 Orwell-Vernon Estuary</b> | <b>NAI</b>                                                                                                                                                                                                                                                                                                                             | <b>MR</b>   | <b>MR</b>   |
| Rationale                         | Given the initially low risk levels the proposed strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets while preserving natural shorelines. This will include raising roads as required and allowing bridges and culverts to allow full tidal passage. |             |             |
| <b>7.01 Brush Wharf -PU</b>       | <b>R</b>                                                                                                                                                                                                                                                                                                                               | <b>R</b>    | <b>R</b>    |
| Rationale                         | A Resist strategy is proposed for this key infrastructure. Future works would also require flood mitigation, subject to final strategies to be developed with the local stakeholders.                                                                                                                                                  |             |             |

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.